

Multi-agent Deep Reinforcement Learning collaborative Traffic Signal Control method considering intersection heterogeneity[☆]

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ABSTRACT

Traffic Signal Control (TSC) plays a crucial role in mitigating congestion. The extensive integration of Deep Reinforcement Learning (DRL) into TSC, fueled by the advancements in artificial intelligence, has yielded remarkable efficacy. Nevertheless, prevailing DRL models are typically developed within homogeneous road networks, characterized by identical intersection types and relatively balanced traffic flow patterns. This predominant approach often results in an oversight of the impact of intersection heterogeneity, encompassing variations in intersection geometric structure, traffic demands, signal timing design, and other pertinent factors, on action selections by researchers. This paper addresses the aforementioned gap by introducing a value-decomposition-based spatiotemporal graph attention multi-agent DRL model (MARL SGAT) that expressly accommodates intersection heterogeneity. Specifically, a heterogeneous correlation index is designed using road network structural parameters and serves to formulate a heterogeneous reward function, enabling the quantification of action execution improvements in diverse road network environments. To address the intricate spatiotemporal features of traffic flows within the heterogeneous road network, the model strategically selects and incorporates these features during the formulation by leveraging a spatiotemporal graph attention network. On these bases, double dual networks are introduced to convert the individual-global-max constraint of action selection into a value range constraint of the action advantage function, thereby facilitating model learning. Simulation results showcase the superior performance of the MARL SGAT algorithm when compared to seven baseline algorithms. Specifically, the algorithm manifests in reduced average vehicle delays, decreased number of stops, and elevated travel speeds within the controlled road network. These advantages are particularly conspicuous in heterogeneous road network environments.

1. Introduction

Traffic signal control (TSC) is pivotal for enhancing intersection traffic efficiency, mitigating safety risks, and reducing vehicle travel costs. As a frequently employed method, collaborative control views adjacent intersections holistically, formulating an optimal signal control plan through systematic adjustments to signal cycle length, green time, and other parameters. Traditional signal control approaches fall into three categories based on control principles: fixed-time control, actuated control, and adaptive control. Among

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these, adaptive control stands out as it can achieve and maintain a global performance level for the control system by accommodating fluctuations in traffic parameters over time. It serves as a prevalent approach to address signal control challenges.

In recent years, the integration of machine learning methods into TSC has become increasingly prevalent with the advent of artificial intelligence (Qin et al., 2022; Qu et al., 2023). Reinforcement learning (RL), a prominent machine learning algorithm, finds frequent application in adaptive traffic signal control due to its merits in organizational optimization and real-time decision-making (Joo et al., 2020; Acar and Sterling, 2023). In RL, the agent continually updates the mapping relationship, i.e., the signal control policy, between actions and environmental states through interactions with the environment. It then adaptively adjusts the signal control scheme based on the long-term cumulative reward value (Q value) derived from the adopted policy. However, challenges arise in processing high-dimensional continuous state information, rendering the algorithm less effective in complex road network environments. The emergence of deep reinforcement learning (DRL) offers a solution to these issues. Leveraging the potent perception processing capabilities of deep learning (DL) (Nigam and Srivastava, 2023), the agent autonomously observes and learns the state representation of the road network environment. It undergoes decision-making training based on RL, optimizing problem-solving strategies and methods, ultimately achieving end-to-end (from input to output) control and perception. Consequently, traffic signal control methods grounded in DRL algorithms have garnered increasing attention among scholars (Liang et al., 2020; He et al., 2024).

In the early stages, DRL applied to TSC mainly focused on isolated intersections. Subsequently, methodologies such as Q learning (Genders and Razavi, 2019), SARSA (Kekuda et al., 2021), or a hybrid approach combining fuzzy control and RL (Kumar et al., 2020), were progressively expanded to encompass the signal control of multiple signalized intersections. During this phase, agent actions in the algorithm were primarily influenced by their local environmental states, with limited information exchange between agents. Consequently, achieving the collaborative control objective proved to be challenging. Addressing the challenge of collaborative control for multiple signalized intersections necessitates understanding the interactive relationships between intersections within the road network, which is crucial for formulating optimal control policy (Li et al., 2021; Tang and Zeng, 2022). To tackle this, certain scholars have employed multi-agent reinforcement learning (MARL) algorithms (Liu et al., 2023; Su et al., 2023; Fei et al., 2024). By facilitating agent communication (Song et al., 2024) or employing value decomposition (Wang et al., 2021a) to share the environment, these approaches adjust the signal control policy of each intersection from a network-wide perspective. Wang et al. (2021b) leveraged the cooperative vehicle infrastructure system to construct a MARL framework predicated on collaborative groups. This framework enhanced the communication among signal light agents through a k-nearest neighbor state representation and a spatially influenced discounted joint reward model, thereby fostering cooperation. In 2023, Yu et al. developed an RL-based decentralized controller that integrated bus priority with TSC to optimize the efficiency of multimodal transportation networks. This method achieved communication and collaboration among multiple agents by sharing the action information of the agents. Yazdani et al. (2023) proposed an adaptive traffic signal model based on DRL focusing on optimizing the management of pedestrian and vehicle flows. By designing an extended reward function, the model accurately captures various interactive delays between vehicles and pedestrians, significantly improving the efficiency and effectiveness of traffic control. In the same year, Su et al. developed a multi-agent reinforcement learning (MARL) framework that incorporated sharing strategies and spatial discount factors, effectively decoupling the complex relationship between emergency vehicle navigation and traffic signal control. Chu et al. (2019) added the environment state of adjacent agents to the state expression of the local agent, so that the local agent had more information about regional traffic distribution and cooperation strategies, and improved the fitting ability of the model. Chen et al. (2021) proposed a decentralized MARL method based on participant evaluation to control traffic signals. They employed a differential reward method to assess each agent's contribution within the cooperative game, which expedited the convergence speed of the algorithm. Considering the importance of information sharing in large-scale road networks, Jiang et al. (2022) designed a general communication form UniComm, which embedded a large number of observations collected on an agent into crucial predictions of the impact on its adjacent agents, improving network communication efficiency.

Moreover, the exploration and incorporation of spatiotemporal correlations among traffic flows have become crucial in the development of machine learning-based signal control models. The traffic flow system is a highly complex and dynamically evolving entity. Variations in traffic flow patterns within individual road sections are not only influenced by their unique historical trends but are also intricately connected to the traffic dynamics in other road sections. This implies the presence of factors that impact traffic flow changes across both temporal and spatial dimensions. Consequently, an effective traffic control model must possess an input structure that accommodates spatiotemporal characteristics simultaneously (Dong et al., 2021; Wang et al., 2022). Notably, Huang et al. (2021) introduced a MARL model, leveraging a spatial-temporal attentive neural network, to optimize traffic signal timing in a comprehensive road network. Zhao et al. (2019) used the graph convolutional networks (GCN) to learn complex topological structures for capturing spatial dependencies. The gated recurrent unit was employed to discern dynamic changes in traffic data, thereby capturing temporal dependencies. Wang et al. (2020b) realized the coordinated control of multi-intersection in a distributed manner through deep q-network (DQN) (Mnih et al., 2013). In this model, the traffic light adjacency graph was constructed based on the spatial structure among traffic lights, and the historical traffic records would be integrated with the current traffic state via a recurrent neural network structure. Wei et al. (2019) proposed a model, CoLight, to enable cooperation of traffic signals, which used graph attention networks (GAT) (Veličković et al., 2017) to facilitate communication. For a target intersection in a network, CoLight could not only incorporate the temporal and spatial influences of neighboring intersections to the target intersection but also build up an index-free model of neighboring intersections. Wang et al. (2024) used GAT to capture geometric relationships between lanes and utilized an enhanced GraphSAGE technique to explore the spatial characteristics of the traffic network. This method improved the model's adaptability to various traffic conditions and road networks, allowing for flexible traffic signal control adapted to the

geometric shapes of different intersections. Wang et al. (2022) designed a graph neural framework with GAT and long short-term memory (LSTM) networks to obtain spatial and temporal information for coordinated control of traffic signals.

In conclusion, existing research approaches the problem from multiple perspectives such as feature extraction, reward reconstruction, and model structure optimization. These approaches employ a range of DL methods to uncover the spatiotemporal correlation between traffic flows and implement collaborative control of multiple signalized intersections based on RL methods. However, practical scenarios often involve a common urban road network structure that includes sub-arterial or branch roads connecting two arterial roads. This leads to significant variations in the geometric structure, traffic demand, and signal timing design among adjacent intersections, collectively termed “intersection heterogeneity” for this discussion. In a heterogeneous road network, traffic conditions at each intersection markedly differ from those in a homogeneous environment, characterized by similar intersection types and relatively balanced traffic flow. Two notable scenarios arise: (i) the traffic stream from a larger upstream intersection to a smaller downstream intersection is divided by red lights, causing vehicles at the end of the traffic stream to wait at the intersection stop line until the next green phase, significantly increasing travel time; (ii) the traffic stream from a smaller intersection to a larger one intermittently experiences a “stop-and-go” condition during the green phase. Frequent stops and starts reduce average vehicle speed, leading to delays and increased energy consumption (Ji et al., 2024; Ma et al., 2023). The existence of heterogeneous intersections amplifies the uncertainty and nonlinearity of traffic flow in the road network, complicating spatiotemporal correlations among traffic flows. Despite the inherent complexity, most models still construct the reward function and verify the model based on homogeneous road networks, overlooking the influence of intersection heterogeneity on action selection. Consequently, these models are unsuitable for heterogeneous road network environments with significant differences in intersection types.

To address the aforementioned issues, this study uniquely accounts for intersection heterogeneity in the collaborative control problem, by formulating a value decomposition-based spatiotemporal graph attention multi-agent deep reinforcement learning model, termed MARL_SGAT. The objective of this model is to maximize the long-term cumulative reward across the entire road network. This model also involves treating the adjustment of green time at individual intersections as actions. The primary contributions of this work can be summarized in three key aspects:

- We introduce a MARL framework for collaborative traffic signal control where each intersection is managed by an individual agent. This framework accommodates intersection heterogeneity depicted by relevant spatiotemporal features, leveraging the spatiotemporal GAT for extraction. Additionally, the framework employs the joint action-value function decomposition, strategically reducing state–action space and encouraging collaborative interactions among multiple agents for mutual benefit. Consequently, our framework exhibits the capability to adapt the signal control policy of individual intersections, offering a holistic network viewpoint.
- Considering the prevalent distinctions among adjacent intersections in urban road networks, we formulate a heterogeneous correlation index to quantify variations in intersection topology, providing a robust foundation for our framework. Employing this index, a novel reward function is devised to offer a more precise evaluation method for the agent’s action selection within a heterogeneous road network environment.
- The double dual networks that we introduced decompose the action-value function for both the global road network and individual intersections into the sum of the state value function and the advantage function. This alteration detaches the state value function from complete dependency on the action, thereby expediting the convergence speed of the network. Simultaneously, it transforms the full IGM constraints into value range constraints for the advantage function, offering a more straightforward implementation. This transformation simplifies the learning process for the optimal action-value function.

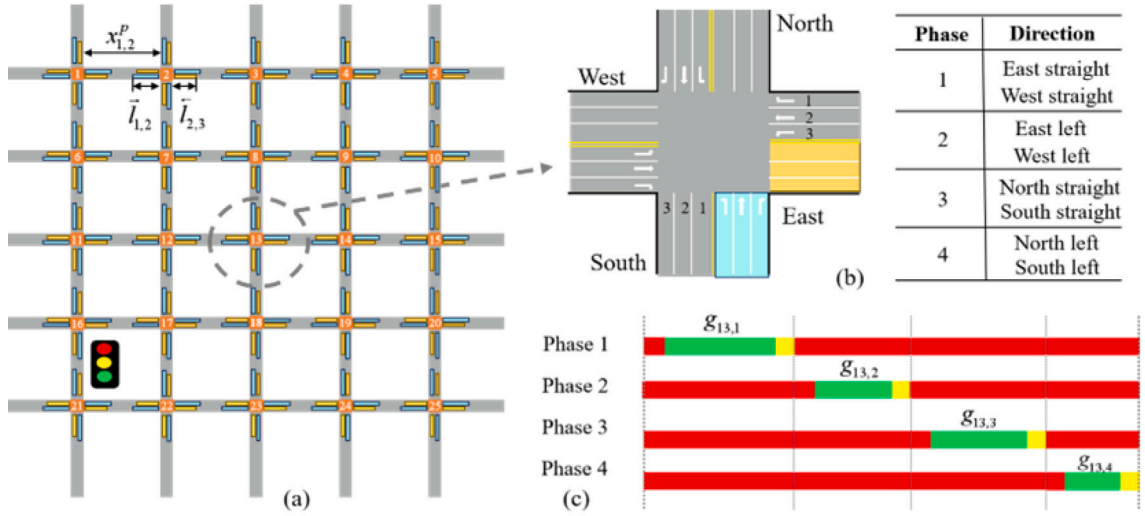
The subsequent sections of this paper are structured as follows. In Section 2, we present the fundamental concepts of environment, action, state, and reward within the MARL model established in this study. Section 3 provides a detailed description of the model framework and structure. In Section 4, a simulation-based case study is presented, and the proposed method is compared with other baseline methods. The concluding section, Section 5, summarizes the entire paper and draws conclusions.

2. Agent formulation

2.1. Environment and action

In the model presented in this paper, each intersection is managed by an individual agent. Intersections and the corresponding agents are denoted by the same set of indices i ($i = [1, 2, \dots, I]$). I is the number of intersections in the road network. Intersection i is characterized as a J_i -way intersection, with each entrance comprising $\bar{Z}_j$ lanes and each exit having $\bar{Z}_j$ lanes. The entrances and exits refer to the passages in the intersection that are specifically used for vehicles to enter or leave the road. For ease of reference, each entrance and exit are sequentially numbered in a clockwise direction. The lengths of the entrance and exit are denoted as $\bar{l}_{i,j}$ and $\bar{l}_{i,j}$ ($j = [1, 2, \dots, J_i]$), respectively. Intersection i has P_i phases, each having a green time of $g_{i,p}$ ($p = [1, 2, \dots, P_i]$), and the phase sequence at each intersection remains constant. Fig. 1 shows the structure of a 5×5 traffic network (composed of 25 homogeneous intersections), that is, $I = 25$. For the convenience of description, we have numbered the intersections. Take Intersection 13 as an example. This intersection is a four-leg intersection, $J_{13} = 4$. Its entrance road and exit road correspond to the blue and yellow box lines in the figure respectively, and both have three lanes. The signal timing scheme of this intersection is shown in Fig. 1(c). Specifically, the intersection has four signal phases, $P_{13} = 4$, and the green time of each phase is $g_{13,1}, g_{13,2}, g_{13,3}, g_{13,4}$.

The action undertaken by the agent is defined as a_i^t . In the context of the TSC task, actions refer to the adjustments made by the agent to the signal lights. Action settings usually contain three approaches: phase selection (He et al., 2021; Wang et al., 2022;

Fig. 1. Example of the 5×5 multi-intersection scenario.

Kong et al., 2022; Bokade et al., 2023), phase shift (Kumar et al., 2020; Liu et al., 2023; Wei et al., 2018), and phase duration adjustment (Li et al., 2021, 2020; Yoon et al., 2021; Luo et al., 2024). Phase selection involves choosing the appropriate phase from a predefined set, offering flexibility in the combination of phases at the cost of relatively frequent changes in signal light colors. On the other hand, phase shift is to determine whether to maintain the current phase, and phase duration adjustment involves adjusting the green time based on specific rules, which are similar. In contrast to phase selection, the sequence of phase changes under the two latter methods is relatively predetermined, aligning more closely with driver expectations (Mao et al., 2022). Therefore, we select phase duration adjustment as the action-setting method concerning driver experience and preferences. Specifically, the agent has three candidate operations: (i) maintaining the green time of the current phase; (ii) extending the green time of the current phase by $\Delta \tilde{r}$ seconds; (iii) reducing the green time of the current green phase by $\Delta \tilde{r}$ seconds. On this basis, we formulate the following rules by setting the minimum green time $g_{i,min}$ and the maximum green time $g_{i,max}$ (Zhang and Wang, 2010): the agent executes the recommended action if the current phase's green time is greater than $g_{i,min}$ and the duration will not exceed $g_{i,max}$ after this action. Otherwise, the current recommended action is ignored, and the agent awaits the next moment ($t + \Delta t$) for reassessment and adjustment.

2.2. State

In the multi-intersection cooperative signal control model based on MARL, the state includes two parts: the individual state of the intersection and the global state of the road network. For individual states, the traffic condition of each lane within an intersection is typically defined by a combination of available parameters such as the count of vehicles, the number of queued vehicles, and phase information (Wei et al., 2021; Wang et al., 2021c). However, owing to spatial structure constraints in the road network, an identical count of vehicles and queued vehicles on different lanes within the same intersection may correspond to distinct traffic conditions. Therefore, road spatial structure should be captured as well to depict various traffic conditions on each lane at intersections. Here we utilize road network structure parameters to reflect intersection state characteristics. Specifically, the instantaneous state parameter s_i^t of the intersection at time t consists of the entrance state collection $\vec{s}_i^t$, the exit state collection $\vec{s}_i^t$, and signal states p_i^t ($p_i^t = 1, 2, \dots, P_i$), as shown in Eq. (1):

$$s_i^t = \{\vec{s}_i^t, \vec{s}_i^t, p_i^t\} \quad (1)$$

Among $\vec{s}_i^t$, the traffic state of lane z at entrance j comprises the lane's traffic density $\tilde{k}_{i,j,z}^t$ and queuing density $\tilde{K}_{i,j,z}^t$ ($z = [1, 2, \dots, \tilde{Z}_i]$), which are calculated as shown in the Eqs.:

$$\tilde{k}_{i,j,z}^t = \frac{\tilde{n}_{i,j,z}^t}{\tilde{l}_{i,j}} \quad (2)$$

$$\tilde{K}_{i,j,z}^t = \frac{\tilde{N}_{i,j,z}^t}{\tilde{l}_{i,j}} \quad (3)$$

where $\tilde{n}_{i,j,z}^t$ and $\tilde{N}_{i,j,z}^t$ are respectively the counts of vehicles and queued vehicles on lane z in intersection i 's entrance j at time t , pcu.

Since vehicles usually pass at free-flow speeds through exit lanes, whether those lanes contain sufficient space deserves more attention than the traffic density and queuing density there. Hence we use the remaining space of exit lane z , denoted as $\bar{c}_{i,j,z}^t$ to express the exit's state of lane z at entrance j , which is also a fundamental element in $\bar{s}_i^t$. The $\bar{c}_{i,j,z}^t$ is represented by the ratio of the count of vehicles on lane z in intersection i 's entrance j at time t , $\bar{n}_{i,j,z}^t$, to the maximum number of vehicles that the lane can accommodate, $\bar{n}_{i,j,z}^{\max}$ ($z = [1, 2, \dots, \bar{z}_i]$). The value of $\bar{n}_{i,j,z}^{\max}$ is equal to the ratio of the entrance road length $\bar{l}_{i,j}$ to the standard car length l_v .

$$\bar{c}_{i,j,z}^t = 1 - \frac{\bar{n}_{i,j,z}^t}{\bar{n}_{i,j,z}^{\max}} \quad (4)$$

Regarding road network traffic state S^t , some studies directly concatenate all the state information from each intersection to indicate the global information of the road network (Liu et al., 2022; Zhang et al., 2019; Ma and Wu, 2022). While this approach accurately reflects the road network traffic state, it also results in an exponential increase in the dimensions of the traffic state features as the number of intersections rises. This dilemma poses a considerable challenge to network training. Nevertheless, effectively conveying the overall traffic state information of large-scale road networks with fewer variables becomes a crucial task in the collaborative control of heterogeneous intersections. In this study, we select representative feature parameters, including the average values of traffic density and queuing density of each entrance at the intersection and the phase number (p) to simplify the representation of individual intersection traffic states, and the number of vehicles entering and exiting the road network during period $[t - \Delta t, t]$ (n_{in}^t and n_{out}^t) as the global state expression. Incorporating these features, as shown in Eq. (5), not only efficiently reflects the traffic state of individual intersections but also signifies the traffic characteristics of the entire road network.

$$S^t = \{s_1^t, s_2^t, \dots, s_I^t, n_{in}^t, n_{out}^t\} \quad (5)$$

2.3. Heterogeneous correlation index and reward

In MARL algorithms, rewards should possess spatial decomposability and be easily measurable following an agent's action (Ma et al., 2021). As stated in the Introduction, the incorporation of intersection structure information into the reward function calculation process is necessary, with the heterogeneity between intersections primarily indicated by variations in lane channelization and signal phase design. Accordingly, we utilize the structural information of the target intersection i and its upstream and downstream intersections to formulate an intersection heterogeneous correlation index. The heterogeneous correlation between intersection i and the upstream intersection i^+ of the critical lane in phase p and the heterogeneous correlation between the intersection i and the downstream intersection i^- of the critical lane in phase p , $\bar{cor}_{i,i^+}^p$ and $\bar{cor}_{i,i^-}^p$, are calculated as follows:

$$\bar{cor}_{i,i^+}^p = \frac{\bar{m}_{i^+,i}^p}{m_i^p \sqrt{x_{i^+,i}^p}} \quad (6)$$

$$\bar{cor}_{i,i^-}^p = \frac{\bar{m}_{i,i^-}^p}{m_i^p \sqrt{x_{i,i^-}^p}} \quad (7)$$

where m_i^p is the lane count corresponding to the critical traffic flow f_i^p in phase p of intersection i ; $\bar{m}_{i^+,i}^p$ is the lane count corresponding to the traffic flow merging into f_i^p at the upstream intersection i^+ ; $\bar{m}_{i,i^-}^p$ is the lane count of exits on the downstream intersection i^- that f_i^p heads towards; $x_{i^+,i}^p$ is the distance between intersection i and the stop line of the upstream intersection i^+ , m; x_{i,i^-}^p is the distance between intersection i and the stop line of downstream intersection i^- , m; q is the scaling factor used for unifying variables' numerical range.

Furthermore, the control objective of traffic signals varies with alterations in traffic supply and demand and is reflected by the shift of different reward functions. During off-peak periods, when traffic supply adequately meets demand, traffic signal control aims at reducing traffic delays and enhancing traffic efficiency. Hence, the average vehicle delay at individual intersections is selected as the feature parameter to measure the action's impact. Conversely, during peak periods with heightened traffic demand, certain intersections may experience oversaturated traffic conditions. During such times, the primary objective of signal control is to increase the throughput of the intersection. Therefore, the throughput of the intersection and the average vehicle delay serve together as the selected feature parameters for assessing the action's impact during peak hours. On this basis, considering the prevalent imbalanced traffic demand of urban road networks, we utilize the ratio of the hourly volume $O_i^{u,p}$ of the critical lane in phase p of intersection i to the hourly traffic volume O_i^u of the intersection under varying traffic demands as the weighting factor ($u = 0, 1$, corresponds to off-peak and peak hours respectively), as expressed as Eq. (8). This factor is employed to calculate the reward r_i^t that agent i can attain by executing action a_i^t at time t .

$$r_i^t = \begin{cases} \sum_{p=1}^{P_i} \frac{O_i^{u,p} \bar{cor}_{i,i^+}^p + \bar{cor}_{i,i^-}^p}{O_i^u} d_i^{t,u,p}, & \text{if } u = 0 \\ \sum_{p=1}^{P_i} \frac{O_i^{u,p} \bar{cor}_{i,i^+}^p + \bar{cor}_{i,i^-}^p}{O_i^u} (w_{i,D}^t d_i^{t,u,p} + w_{i,O}^t o_i^{t,p}), & \text{if } u = 1 \end{cases} \quad (8)$$

where $d_i^{t,u,p}$ is the average vehicle delay of f_i^p during the unit time step Δt starting from time t in period u . Its value is equal to the quotient of the total stop time and the number of stopped vehicles of the traffic flow f_i^p ; s_i^t represents the throughput, namely

the number of vehicles leaving the stop line, of f_i^p during the unit time step Δt starting from time t in peak hours, pcu; $w_{i,D}^t$ and $w_{i,O}^t$ are respectively the weight values corresponding to the average vehicle delay and traffic volume, where $w_{i,D}^t$ is negative and $w_{i,O}^t$ is positive.

Global reward R^t is the aggregate of rewards across all intersections as follows:

$$R^t = \sum_{i=1}^I r_i^t \quad (9)$$

The long-term cumulative reward Q_i^t of agent i in state s_i^t is expressed as Eq. (10). Q_i^t is the action-value function that requires deep network fitting in deep reinforcement learning.

$$Q_i^t = E[r_i^t + \gamma r_i^{t+1} + \dots + \gamma^c r_i^{t+c}] \quad (10)$$

where γ is the discount factor to reduce the importance of future rewards and balance current rewards with future rewards, $\gamma = [0, 1)$. γ approaching 0 indicates a prioritization of short-term rewards in action selection, while γ approaching 1 signifies that the agent places greater emphasis on the acquisition of long-term rewards through action execution; c represents the maximum period considered for reward calculations.

Eq. (10) is usually simplified as follows:

$$Q_i^t = E[r_i^t + \gamma \max Q_i^{t+1}] \quad (11)$$

3. Methodology

3.1. Overall framework

This paper introduces a collaborative control model for heterogeneous intersections based on a MARL algorithm. The comprehensive network framework is depicted in Fig. 2. The network takes as input the local environmental state s^t of each agent at time t and the overall traffic state S^t of the road network. The output is the joint action-value Q_{tot}^t encompassing all agents. The calculation of this value is executed through three modules: spatiotemporal feature extraction, individual value function fitting, and global value function calculation.

The spatiotemporal feature extraction module incorporates GAT and gated recurrent unit (GRU) (Cho et al., 2014) as the primary network components. This module takes the local environmental state of each agent as input and generates an output feature matrix containing spatiotemporal information. Subsequently, this matrix serves as one of the input parameters in the subsequent process and is fed into the individual value function fitting module. The blue block diagram of Fig. 2 shows in detail the information flow of agents 1 and I in the spatiotemporal feature extraction module, where (s_1^t, s_{1*}^t) and (s_I^t, s_{I*}^t) represent local states of agents 1 and I , and local states of agents with direct connectedness with 1 and I . H_1^{t-1} and H_I^{t-1} are hidden states of agents 1 and I at $t-1$.

The individual value function fitting module, denoted by the yellow block in the figure, receives input information comprising the spatiotemporal feature matrix and the global state information S^t of the road network. The module outputs the state function value V^t and advantage function value A^t for each agent in the global environment. Within the individual value function fitting module, the spatiotemporal feature matrix is employed to fit the agent's state value function and action advantage function. Simultaneously, the global state information S^t utilized to generate the weight w_{TR}^t and bias b_{TR}^t . The action is determined based on the advantage function value for each agent, obtained through a greedy policy exploration. The diagram in the purple block illustrates the global value function calculation module, primarily responsible for calculating the global (joint) action function value Q_{tot}^t . The input information for this module comprises the action of each intersection, the global state information S^t , and the output results of the individual value function fitting module. The first two serve as input information for the hybrid network within the module, generating the weight of the advantage function value. This weight signifies the importance of each agent in the global network. Details about the component modules are elucidated in subsequent sections.

3.2. Spatiotemporal feature extraction module

Intersection actions are determined by the action function value, and the computation of this value relies on the traffic states of each intersection and the road network. Hence, precise state representation is pivotal for effective action selection. In Section 2.2, we defined the traffic state of the intersection at time t , utilizing traffic density, queuing density, and signal phase as feature parameters to depict the traffic state. While this expression conveys the environmental state of individual intersections, it lacks the correlation of traffic flows between adjacent intersections. To enhance the fitting of the action function value for each intersection, we employ a combined network utilizing the GAT and GRU to further investigate the spatiotemporal correlation between intersections. Specifically, GAT facilitates the selective acquisition of information from adjacent intersections. Taking intersections as graph nodes and road segments between intersections as edges, we simplify the road network into a graph structure composed only of points and lines. The set of adjacent nodes of intersection i is denoted as I_i^* , with intersection i having J_i adjacent nodes. The projection matrix w_i^t is applied here to linearize the state information of node i and its adjacent node i^* , thereby altering the dimensions of the state features. Building upon this, the node features are concatenated in sequence, fed into the attention network denoted as θ ,

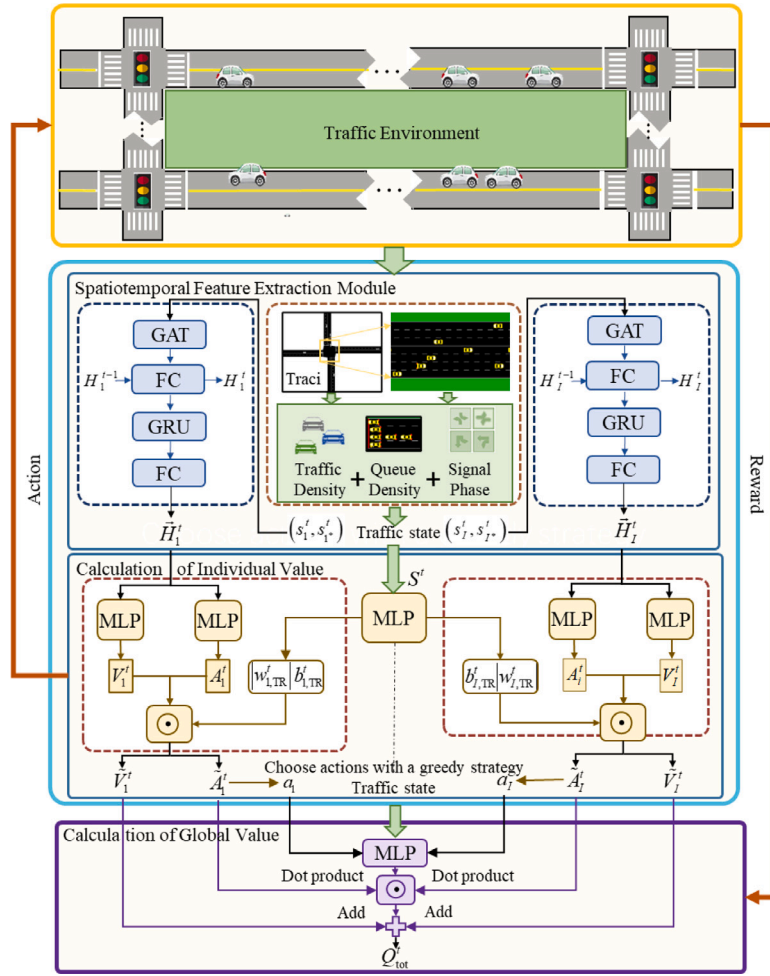


Fig. 2. Model framework.

and projected to a scalar value. Ultimately, by subjecting this scalar value to a nonlinear transformation through the activation unit $\text{LeakyRelu}(\cdot)$, we can derive the correlation coefficient ϕ_{i,i^*}^t between the target intersection and its adjacent intersections at time t :

$$\phi_{i,i^*}^t = \text{LeakyRelu}(\vartheta(w_{i,i^*}^t \parallel w_{i,i^*}^t)) \quad (12)$$

Concatenating operation $\parallel$ leads to $\phi_{i,i^*}^t \neq \phi_{i^*,i}^t$, meaning that attention value is non-symmetrical, which fits the actual situation. In urban road networks, imbalanced traffic demand results in distinct variations in the impact relationship between adjacent intersections and the target intersection across different scenarios. For instance, if there is an increased traffic volume from intersection i to intersection i^* , the influence of intersection i on intersection i^* will be more pronounced, $\phi_{i,i^*}^t > \phi_{i^*,i}^t$, in contrast, ϕ_{i,i^*}^t has a smaller value, $\phi_{i,i^*}^t < \phi_{i^*,i}^t$. Subsequently, the softmax($\cdot$) function is employed to normalize the attention coefficients between node i and its adjacent nodes (as in Eq. (13)).

$$\delta_{i,i^*}^t = \text{softmax}(\phi_{i,i^*}^t) = \frac{\exp(\phi_{i,i^*}^t)}{\sum_{i^* \in I_i^*} \exp(\phi_{i,i^*}^t)} \quad (13)$$

Ultimately, the state information ζ_i^t of intersection i is obtained by combining the state information from each adjacent node and the activation function $\sigma(\cdot)$:

$$\zeta_i^t = \sigma\left(\sum_{i^* \in I_i^*} (\delta_{i,i^*}^t w_{i,i^*}^t s_{i^*}^t)\right) \quad (14)$$

Utilizing the network structure delineated in Eqs. (12)–(14), we selectively incorporate the state information of adjacent intersections at the same moment into the state representation of the target intersection. This approach yields a more comprehensive

state information set under conditions of incomplete observation. Subsequently, we introduce the GRU network to capture the temporal relationships within the intersection state space. This enables the integration of variations and underlying patterns inherent in historical traffic states into the state expression process. Preceding this, a single-layer fully connected neural network is employed to uniformly convert the state information ζ_i^t of intersection i into a high-dimensional vector space (as in Eq. (15)). This facilitates more effective learning and extraction of correlation features between variables.

$$\zeta_{i1}^t = w_{i,\zeta 1}^t \zeta_i^t \quad (15)$$

where $w_{i,\zeta 1}^t$ is the learnable weight factor of a fully connected neural network.

GRU usually consists of two components: the reset gate and the update gate. Notably, these gating mechanisms retain information in long-term sequences without discarding it over time, irrespective of its relevance solely to predictions at the current moment. More precisely, the process involves utilizing the hidden state H_i^{t-1} inherited from the previous moment, representing the feature set containing historical traffic information (with the initial moment H_i^{t-1} set to 0). Additionally, the state information ζ_i^t of the target node, integrated with information from adjacent nodes, serves as input to obtain the reset gate signal τ_i^t and update gate signal η_i^t , as shown in Eqs. (16) and (17).

$$\tau_i^t = \sigma \left(\zeta_{i1}^t w_{i,\zeta\tau}^t + H_i^{t-1} w_{i,h\tau}^t + b_{i,\tau}^t \right) \quad (16)$$

$$\eta_i^t = \sigma \left(\zeta_{i1}^t w_{i,\zeta\eta}^t + H_i^{t-1} w_{i,h\eta}^t + b_{i,\eta}^t \right) \quad (17)$$

where $w_{i,\zeta\tau}^t$, $w_{i,h\tau}^t$, $w_{i,\zeta\eta}^t$ and $w_{i,h\eta}^t$ are learnable weight factors; $b_{i,\tau}^t$ and $b_{i,\eta}^t$ are bias terms; $\sigma(\cdot)$ stands for a nonlinear activation function, being able to transform the data to values ranging from 0 to 1, thereby acting as a gating signal.

When τ_i^t is approaching 0, the model will discard the hidden information in the historical traffic state, leaving only the current input information. Conversely, when τ_i^t is approaching 1, the model will assume that all historical information is valid, and add it to the state information of the intersection at the current moment. Leveraging the reset gate gating signal, we can compel the hidden state to discard any information from the future deemed irrelevant to predictions as outlined in Eqs. (18) and (19).

$$\tilde{H}_i^t = \tanh(\zeta_{i1}^t w_{i,\zeta H}^t + \tau_i^t \odot H_i^{t-1} w_{i,hH}^t + b_{i,H}^t) \quad (18)$$

$$H_i^t = (1 - \eta_i^t) \odot H_i^{t-1} + \eta_i^t \odot \tilde{H}_i^t \quad (19)$$

For the convenience of description, we use the variable w to uniformly represent the weight coefficients appearing in the paper and use the variable b to represent the bias term, with different upper and lower subscripts to distinguish them. Building on this, the update gating signal regulates the degree to which the state information from the previous moment influences the current state, thereby updating the hidden state H_i^t . Subsequently, a fully connected layer is once again employed to alter the vector dimension, ultimately yielding the final output result $\tilde{H}_i^t$:

$$\tilde{H}_i^t = w_{i,H1}^t H_i^t \quad (20)$$

where $\tanh(\cdot)$ represents hyperbolic tangent activation function, for limiting data to the range $[-1, 1]$; $(1 - \eta_i^t) \odot H_i^{t-1}$ stands for the selective forgetting of historical traffic hidden states; $\eta_i^t \odot \tilde{H}_i^t$ is the selective memory of the candidate hidden states of the current node.

3.3. Calculation of individual value

The comprehensive individual value function network is depicted in Fig. 3. The individual value function is synonymous with the individual action-value function, with its value representing the anticipated return that the agent i can achieve by executing action a_i^t in state s_i^t , denoted as Q_i^t . A higher action value for any intersection in the road network is usually obtained from the intersection being in a more favorable traffic condition or through the effectiveness of executing correct actions, i.e., a judicious adjustment of the green time. Therefore, relying solely on individual function values hinders accurate differentiation of the causes behind traffic efficiency improvements. To address this, we employ a dual network to decompose the action-value function $Q_i^t(s_i^t, a_i^t)$ of the intersection into the sum of the state value function $V_i^t(s_i^t)$ and the advantage function $A_i^t(s_i^t, a_i^t)$. The state function value represents the expected return agent i can achieve by following a policy in a given state. Specifically, the state function value equals the average of all action-value functions in a specific state concerning the action probability. The advantage function value represents the value of the current action relative to the average value: if the advantage is greater than zero, it indicates that the action is effective compared to the average action; if the advantage is negative, it implies that the current action is not as effective as the average action in optimizing the environment. The fitting network corresponding to the state value function and advantage function is expressed as Eqs. (21)–(24). The networks are similar in structure, and both take the output of the spatiotemporal graph attention network as input and consist of three fully connected layers. The output yields the state value V_i^t of the intersection at time t and the advantage value A_i^t for each action in this state.

$$a_{i1,V}^t = \text{Relu} \left(\left(\tilde{H}_i^t \parallel a_i^{t-1} \right) w_{i,V1}^t + b_{i,V1}^t \right) \quad (21)$$

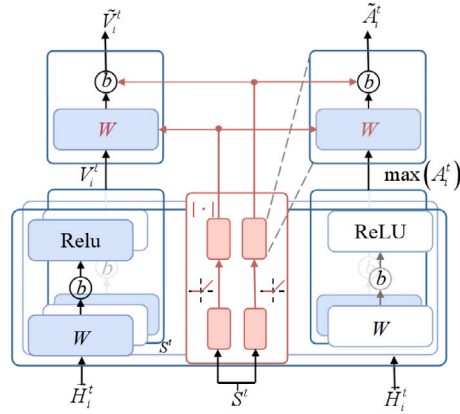


Fig. 3. Structure of individual value function network. The function of the red network in the figure is to generate the weight and deviation values corresponding to the state value function and the advantage value function. The red icons of the network are the absolute value operation and the ReLU activation function from top to bottom.

$$\alpha_{i2,V}^t = \text{Relu} \left(\alpha_{i1,V}^t w_{i,V2}^t + b_{i,V2}^t \right) \quad (22)$$

$$V_i^t = \alpha_{i2,V}^t w_{i,V3}^t + b_{i,V3}^t \quad (23)$$

$$A_i^t = \alpha_{i2,V}^t w_{i,A3}^t + b_{i,A3}^t \quad (24)$$

Through the aforementioned procedure, we acquire the state function value and advantage function value for the individual intersections. Despite utilizing the GAT network to derive state parameters for adjacent intersections, the information available to the agent is still local and constrained. The traffic environments associated with different agents may exhibit significant differences at the same time, thereby augmenting the complexity of calculating the joint action-value function. Therefore, we compute the individual state function value $\tilde{V}_i^t$ and advantage function value $\tilde{A}_i^t$ for each agent in the global environment:

$$\alpha_{i1,\tilde{V}}^t = \text{Relu} \left(\mathbf{S}^t w_{i,\tilde{V}1}^t + b_{i,\tilde{V}1}^t \right) \quad (25)$$

$$w_{i,\text{TR}}^t = \left| \alpha_{i1,\tilde{V}}^t w_{i,\tilde{V}2}^t + b_{i,\tilde{V}2}^t \right| \quad (26)$$

$$\alpha_{i1,\tilde{A}}^t = \text{Relu} \left(\mathbf{S}^t w_{i,\tilde{A}1}^t + b_{i,\tilde{A}1}^t \right) \quad (27)$$

$$b_{i,\text{TR}}^t = \alpha_{i1,\tilde{A}}^t w_{i,\tilde{A}2}^t + b_{i,\tilde{A}2}^t \quad (28)$$

$$\tilde{V}_i^t = w_{i,\text{TR}}^t V_i^t + b_{i,\text{TR}}^t \quad (29)$$

$$\tilde{A}_i^t = w_{i,\text{TR}}^t A_i^t + b_{i,\text{TR}}^t \quad (30)$$

where the value of $w_{i,\text{TR}}^t$ is positive to ensure the monotonicity of the local function and the global function; $\tilde{A}_i^t$ is the advantage value of the best action selected based on the greedy policy, that is, the maximum value in A_i^t .

3.4. Calculation of global value

As shown in Fig. 4, the joint action value Q_{tot}^t is computed based on the output from the individual value function network, combined with the road network's traffic state S^t . The magnitude of the function value indicates the impact of the joint action a_{tot}^t on enhancing the overall state of the road network environment in the current condition. However, in the individual value function network, the selection of agent actions represents the optimal outcome only for individual intersections, which may diverge from the optimal choice in global conditions. To ensure consistency between the two, the model construction must adhere to the IGM constraint (Wang et al., 2020a) as illustrated in Eq. (31).

$$\arg \max A_{tot}^t (S^t, a_{tot}^t) = [\arg \max A_1^t (s_1^2, a_1^t), \dots, \arg \max A_I^t (s_I^2, a_I^t)] \quad (31)$$

To this end, we once again employ a dual network (Wang et al., 2016) to decompose the joint action-value function $Q_{tot}^t (S^t, a_{tot}^t)$ into the sum of the state value function $V_{tot}^t (S^t)$ and the advantage function $A_{tot}^t (S^t, a_{tot}^t)$. This approach aligns with the principles of

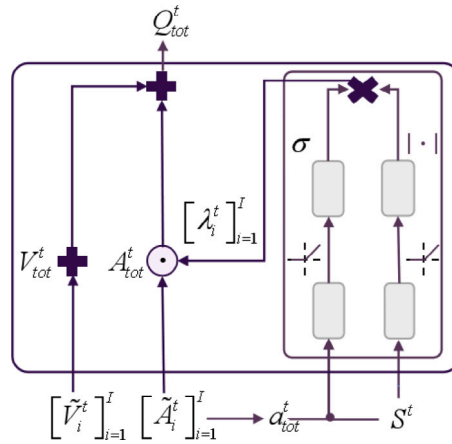


Fig. 4. Mixing network structure. The x-shaped purple icon in the figure represents multiplication, and the meanings of the other icons are consistent with that of Fig. 3.

RL algorithms, where the action-value function and the state-value function become numerically equivalent when the agent executes the optimal action $\tilde{a}_i^t$. Under such circumstances, the individual and global action advantage functions satisfy the relationship illustrated in Eq. (31). Therefore, as Eq. (32), absent other constraints, optimizing the global action advantage function and all individual action advantage functions to attain their maximum values will lead to the same outcome, thereby establishing that the individual optimum aligns with the overall optimum. The consistency constraint of action selection between the entire road network and individual intersections is thus transformed into a value range constraint for the advantage function.

$$A_{tot}^t(S^t, \tilde{a}_{tot}^t) = A_i^t(s_i^t, \tilde{a}_i^t) = 0 \text{ and } A_{tot}^t(S^t, a_{tot}^t) < 0, A_i^t(s_i^t, a_i^t) \leq 0 \quad (32)$$

where $\tilde{a}_{tot}^t$ is the joint optimal action.

At the same moment, for all agents, the overall state of the road network remains constant, and the historical joint state is a predetermined value. Once the policy is established, the value of the intersection state remains unchanged with the actions undertaken by the agent. Accordingly, the sum of the state function values of each agent at time t is directly employed as the state function value V_{tot}^t in the joint state:

$$V_{tot}^t = \sum_{i=1}^I \tilde{V}_i^t \quad (33)$$

The advantage function provides a numerical representation of the value advantage associated with each action within the action set relative to the average level. Its numerical computation relies on the agent's state and the set of available actions in that state. Despite the uniform state of the road network and predetermined optional actions at a given moment, the impact of executing the same action on the road network state at different intersections varies. Consequently, directly computing the cumulative action advantage function value is inappropriate. In this study, leveraging the overall state information S^t of the road network at time t and the collective action a_{tot}^t taken by each intersection, we calculate the significance λ_i^t of each intersection's action adjustment concerning the overall state change of the road network. This is achieved through the hybrid network (Wang et al., 2020a) as follow:

$$\alpha_{i1,\varphi}^t = \text{Relu} \left[(S^t \parallel a_{tot}^t) w_{i,\varphi1}^t + b_{i,\varphi1}^t \right] \quad (34)$$

$$\varphi_i^t = \sigma \left(\alpha_{i1,\varphi}^t w_{i,\varphi2}^t + b_{i,\varphi2}^t \right) \quad (35)$$

$$\alpha_{i1,M}^t = \text{Relu} \left(S^t w_{i,M1}^t + b_{i,M1}^t \right) \quad (36)$$

$$w_{i,M}^t = \left| \alpha_{i1,M}^t w_{i,M2}^t + b_{i,M2}^t \right| \quad (37)$$

$$\lambda_i^t = w_{i,M}^t \varphi_i^t \quad (38)$$

Hence, the advantage function value A_{tot}^t in the joint state is:

$$A_{tot}^t = \sum_{i=1}^I \lambda_i^t \tilde{A}_i^t \quad (39)$$

In the given equation, the activation function σ confines φ_i^t within the interval of 0–1, and taking the absolute value ensures that $w_{i,M}^t > 0$. Therefore, the weight value λ_i^t is a positive value. A positive correlation exists between the advantage function value of an individual intersection and the overall advantage function value of the road network. This further guarantees consistency between action selection and policy.

During the optimization process of the model, if the network update rate is excessively rapid, the agent may struggle to obtain a stable and specific target value, complicating the formulation of a loss function for optimizing the network structure. Therefore, in this study, we establish the loss function based on two network structures – the prediction network and the target network – having the same configuration but different update speeds. Specifically, the prediction network updates parameters in each iteration, while the target network synchronizes the latest weights from the evaluation network every generation to update the network. To make better use of simulation data, we define s_i^t , S^t , a_i^t , s_i^{t+1} , S^t , R^t and the 0–1 variable ρ^t to determine whether the new state is the terminal state as an experience. $\rho^t = 1$ signifies that time $t + \Delta t$ is the terminal state. The experience pool serves as a dynamic repository comprised of experiences. Over time, as the agent iteratively updates its state and receives rewards through ongoing interactions with the environment, the experience pool undergoes continuous updates. Upon reaching a predefined threshold for the volume of data in the experience pool, the agent engages in a playback process, utilizing the records within the pool for dynamic experience. This involves fitting the target label $\hat{Q}_{tot}^t$ (as in Eq. (40)), calculating the loss function $L_{tot}^t(\theta)$ (as in Eq. (41)), and updating the network weights using the gradient descent method.

$$\hat{Q}_{tot}^t = \begin{cases} R^t & \text{if } \rho^t = 1 \\ R^t + \gamma \max \hat{Q}_{tot}^{t+1} & \text{if } \rho^t = 0 \end{cases} \quad (40)$$

$$L_{tot}^t(\theta) = \sum_{\psi=1}^{\tilde{\psi}} (\hat{Q}_{tot}^t - Q_{tot}^t)^2 \quad (41)$$

where $\hat{Q}_{tot}^{t+1}$ is the joint action-value of the target network at time $t + 1$; $\tilde{\psi}$ is the number of data entries randomly selected from the experience pool; θ is the network weight of the prediction network.

Below is the pseudocode of the MARL_SGAT Algorithm. At each time step, the agent obtains a representation of the intersection state s_i^t and feature set of road network state S^t from the environment and relies on the individual value function network to fit the value functions of all optional actions at the intersection in the current state. Subsequently, each agent selects and executes the action a_i^t based on the greedy strategy, and the environment then moves to the next state s_i^{t+1} , and returns the reward r_i^t . After all agents have completed their actions, the model will obtain the overall traffic status S^t of the road network and calculate the overall reward function R^t under the joint action.

4. Experiments

4.1. Experimental setting

In this section, we utilize the traffic simulation software simulation of urban mobility (SUMO) to conduct simulation experiments using both synthetic and real datasets, to evaluate the proposed multi-intersection signal cooperative control model. By invoking the library function in SUMO, we directly retrieve intersection state information, including details such as vehicle count, location, vehicle state, signal phase, and other parameters. The execution of actions and the acquisition of evaluation indicators during model analysis are implemented using the corresponding functions of SUMO. In synthetic and real datasets, each vehicle is denoted as (ID, arr_t, arr_p, lea_t, lea_p), where ID serves as the unique identifier. The remaining variables include the time and location of the vehicle's arrival on the road network and the time and location of its departure. The simulation duration for each round is set to 3600 s for both synthetic and real datasets. Each agent undergoes training for 200 rounds, employing a training data batch size of 32, a discount factor of 0.9, and an experience replay area size of 1000. In addition, according to the hyperparameter tuning results, the time interval Δt between two consecutive actions is set to 12 s; the green light adjustment time Δt is set to 4 s; the value of the discount coefficient q is 3. Other parameters involved in the model are shown in Table 1. For ease of description, we have simplified the expression of the three modules of the spatiotemporal feature extraction module, individual value function fitting module, and global value function calculation module as SFE module, IVFF module, and GVFC module.

Synthetic Dataset: The corresponding road network environment for this dataset encompasses a total of 9 intersections, each configured as a four-way intersection with six lanes. For any given entrance, the lanes contain left turn lanes, straight lanes, and right turn lanes in sequence from the inner to outer direction. The probability of a vehicle taking different directions at the intersection is distributed as follows: 15% for left turns, 75% for straight travel, and 10% for right turns. Each intersection follows a four-phase signal control scheme. The distance between two adjacent intersections is set at 500 m, with a road speed limit of 16.7 m/s. Considering actual traffic conditions, we categorize the daily traffic flow into two periods: off-peak and peak. In these periods, the vehicle arrival rates in each inbound direction (left, straight, and right lanes) are 0.73 pcu/s and 0.97 pcu/s, respectively. In addition, in order to ensure that the data changes in the synthetic data set are more consistent with the real situation, we adjust the vehicle arrival rate to a certain extent during the training process of the model. Specifically, the agent will determine the data type (peak hour data/off-peak hour data) in each round of the simulation experiment based on a Bernoulli random variable (with a value of 0 or 1). On this basis, every ten generations, the agent will generate a random variable between [-0.05, 0.05], and adjust the vehicle arrival rate of the road network based on this variable until the end of training. An example of adjusting the vehicle arrival rate is as follows: if the random number generated at the beginning of the 10th training is -0.03, then during the 10–20

Algorithm 1 MARL_SGAT Algorithm**Input:** Episode length M , Time length T , Discount factor γ , Renewal frequency m .**Output:** Parameter θ .

```

1: Initialize predict  $Q$  network with random weights  $\theta$ .
2: Initialize target  $Q$  network with random weights  $\hat{\theta}$ .
3: for  $episode = 1 \rightarrow M$  do
4:   for  $t = 1 \rightarrow T$  do
5:     Get global state  $S^t$ .
6:     for  $i = 1 \rightarrow I$  do
7:       Get individual state  $s_i^t$ .
8:       if  $random < \varepsilon$  then
9:         With probability  $\varepsilon$  select action  $a_i^t$ .
10:      else if  $random \geq \varepsilon$  then
11:        Sample action from policy  $\pi_i$ .
12:      end if
13:      Execute action  $a_i^t$ .
14:      get reward  $r_i^t$  and next state  $s_i^{t+1}$ .
15:      Add  $(s_i^t, a_i^t, s_i^{t+1})$  to experience  $e$ .
16:       $s_i^t \leftarrow s_i^{t+1}$ 
17:    end for
18:    Calculate global reward  $R^t$ .
19:    Get next global state  $S^{t+1}$ .
20:    Add  $(S^t, R^t, S^{t+1}, \rho^t)$  to experience  $e$ .
21:    Store  $e$  into the experience pool  $E$ .
22:     $S_i^t \leftarrow S_i^{t+1}$ 
23:     $t \leftarrow t + 1$ 
24:  end for
25:  Sample random mini-batch of transition from  $E$ .
26:  Calculate loss  $L_{tot}^t(\theta)$  and update parameter  $\theta$ .
27:  if  $episode/m = 0$  then
28:     $\hat{\theta} \leftarrow \theta$ 
29:  end if
30: end for

```

Table 1
Parameters of MARL_SGAT.

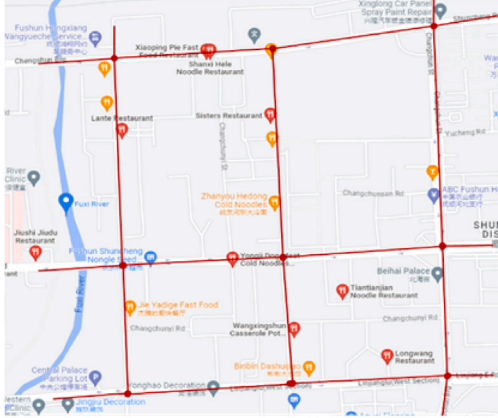
Module	Network	Input size	Output size	Hidden size
SFE module	GAT	9×16	9×32	–
	FC1	1×288	1×576	–
	GRU	$9 \times 64, 9 \times 64$	9×64	9×64
	FC2	1×576	1×288	–
IVFF module	Individual state value	$9 \times 32, 9 \times 1$	9×1	$9 \times 128, 9 \times 32$
	Individual advantage value	$9 \times 32, 9 \times 1$	9×3	$9 \times 128, 9 \times 32$
	Global-individual	$9 \times 38, 9 \times 1$	9×1	$1 \times 64, 1 \times 64$
	State value	$9 \times 38, 9 \times 1$	9×1	$1 \times 64, 1 \times 64$
	Global-individual advantage value	$9 \times 38, 9 \times 1$	9×1	$1 \times 64, 1 \times 64$
GVFC module	Global state value	9×1	1	–
	Global advantage value	$1 \times 38, 1 \times 9, 9 \times 64$	1	$1 \times 64, 1 \times 64$

iterations, the vehicle arrival rate during peak hours of the road network will become $(1-0.03)*0.73 = 0.71$ pcu/s; the vehicle arrival rate during off-peak hours will become $(1-0.03)*0.97 = 0.94$ pcu/s. The above data adjustment methods help simulate situations under different traffic flow conditions so that the model can better adapt to different traffic scenarios and enhance the training performance of the model.

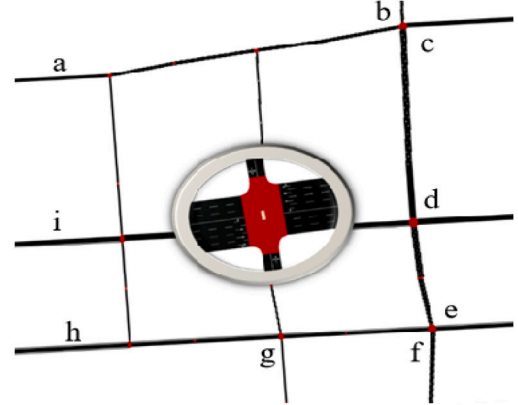
Real Dataset: Satellite images and simulation images depicting actual traffic scenarios were utilized (Fig. 5). Leveraging monitoring equipment in this region, we conduct traffic flow counts at the specified intersections throughout both peak (6:30–8:30) and off-peak periods (14:00–16:00) over one month. The gathered data is then compiled into a real dataset. Table 2 shows the basic environmental information of each intersection, such as the number of signal phases, the number of lanes, the length of the signal cycle, and the overall vehicle arrival status of the road network. The distribution of vehicles has been added to the supplementary file in the form of a table (letters a to i are entrance numbers in the road network).

Table 2
Real environment settings.

Parameters	Description	Value
I	Number of intersections in the road network	9
$w_{i,D}^t$	Weight value corresponding to the average vehicle delay	-0.1
$w_{i,O}^t$	Weight value corresponding to the traffic volume	1.0
P	Set of intersection signal phases	3, 3, 4, 4, 4, 4, 3, 4, 4
M	Set of intersection lanes	10, 10, 18, 18, 18, 24, 15, 16, 22
C_o	Set of the initial cycle lengths of intersections during off-peak hours (s)	70, 83, 122, 112, 114, 131, 82, 95, 134
C_p	Set of initial cycle lengths of intersections during peak hours (s)	75, 87, 123, 99, 116, 129, 98, 110, 148
Arrival rate (Mean)	Average vehicle arrival rate of road network (pcu/300 s)	103.55, 177.20
Arrival rate (SD)	Standard deviation of vehicle arrival rate in road network (pcu/300 s)	15.15, 48.92



(a) Satellite image of traffic networks.



(b) Simulation traffic networks in SUMO.

Fig. 5. Real traffic scene map.

4.2. Comparison scheme and evaluation metric

To demonstrate the effectiveness of the proposed algorithm, we compare MARL_SGAT with the following baseline algorithms, including traditional algorithms and RL algorithms.

FixedTime: The traffic signal controller operates continuously based on predetermined traffic signal schemes, offering advantages such as cost-effectiveness and easy maintenance. Presently, it is a widely employed method of intersection signal control. In this study, the timing schemes for each intersection within both the synthetic and real road networks during various periods were computed using the Webster algorithm.

IQL: Each intersection is autonomously governed by an individual intelligent agent, and there exists no communication mechanism among these agents. Actions are solely chosen based on the holistic environmental conditions to realize cooperative control of multiple signalized intersections.

DDQN-PER: The agent controls the traffic signals of the regional road network through the DQN network with an experienced playback mechanism, and defines the intersection environmental state and reward function simply and consistently during the model construction process (Bouktif et al., 2023).

MGMQ: The multi-layer graph computing method is introduced into DDQN, and GAT and an improved GraphSAGE are used to extract the geometric and spatial characteristics of intersection individuals and intersection road networks; each intersection is controlled by a separate agent, and all agents share a set of parameters (Wang et al., 2024).

CoLight: The intelligent agent realizes collaborative control of regional road network traffic signals based on the combined network of DQN and GAT. On this basis, the attention mechanism is introduced into the network structure to promote agent communication (Wei et al., 2019).

MN_Light: The agent mines the temporal characteristics of historical traffic flow status and action information by adding bidirectional LSTM to DQN to improve the control performance of the model (Kong et al., 2022).

QMIX: Considering the collaborative and competitive relationships among intersections, individual intelligent agents are assigned to control each intersection. Leveraging a blending network, the joint action function value is assessed as an intricate nonlinear combination of each agent's action function values, with structural constraints imposed to maintain consistency between global and individual policies (Rashid et al., 2020).

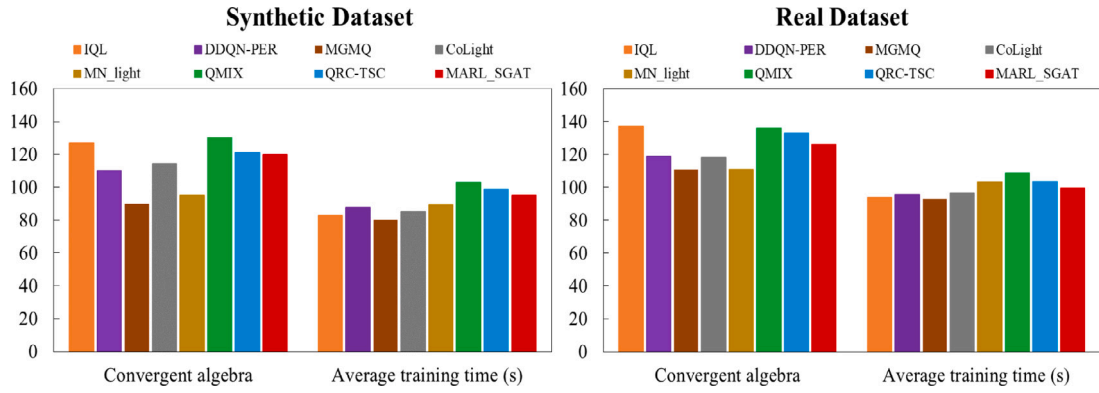


Fig. 6. Comparison of algorithm convergence.

Table 3

Performance of algorithms on synthetic data and real-world data.

Algorithm	Synthetic dataset				Real dataset			
	Delay/s	Travel speed (m/s)	Stops	Throughput pcu	Delay/s	Travel speed (m/s)	Stops	Throughput (pcu)
1 FixedTime	74.55	9.17	2.36	5612.00	124.53	9.69	2.88	3807.00
2 IQL	74.29 ± 0.68	9.33 ± 0.08	2.24 ± 0.09	5633 ± 34.98	98.07 ± 6.75	9.55 ± 0.06	2.21 ± 0.09	3926 ± 46.53
3 DDQN-PER	74.33 ± 1.06	9.39 ± 0.19	2.15 ± 0.17	5644 ± 18.75	100.42 ± 7.85	9.52 ± 0.12	2.24 ± 0.09	3927 ± 61.87
4 MGMQ	72.51 ± 0.77	9.48 ± 0.09	2.03 ± 0.15	5679 ± 32.25	95.03 ± 6.11	9.71 ± 0.09	2.23 ± 0.03	3940 ± 50.17
5 CoLight	73.80 ± 0.60	9.41 ± 0.09	2.12 ± 0.12	5657 ± 24.46	96.12 ± 5.64	9.72 ± 0.09	2.21±0.14	3933±69.78
6 MN_Llight	74.35 ± 0.71	9.37 ± 0.10	2.15 ± 0.07	5643 ± 30.51	97.79 ± 5.23	9.65 ± 0.11	2.20±0.10	3929 ± 47.91
7 QMIX	73.71 ± 0.77	9.42 ± 0.18	2.03 ± 0.03	5696 ± 28.24	95.22 ± 3.60	9.66 ± 0.06	2.33 ± 0.06	3943 ± 31.00
8 QRC-TSC	71.25 ± 2.07	9.51 ± 0.26	2.01 ± 0.11	5714 ± 20.96	94.69 ± 5.27	9.79 ± 0.07	2.24 ± 0.12	3960 ± 31.33
9 MARL_SGAT	70.72 ± 0.93	9.77 ± 0.12	1.89 ± 0.08	5730 ± 6.58	89.68 ± 3.74	10.33 ± 0.05	2.07 ± 0.08	4016 ± 22.30

QRC-TSC: The same processing method as QMIX is adopted to ensure the consistency of global-individual policies. Each intersection is equipped with a separate communication network and can selectively exchange policy messages of variable lengths (Bokade et al., 2023).

4.3. Simulation results and discussion

4.3.1. Contrast experiment

In order to ensure the reliability of the results, we conduct 10 test experiments using different random seeds after the model converged, and the results are summarized in Table 3, with algorithms being sequentially numbered from 1 for clarity. Notably, the DRL-based control models (models 2–9) exhibit superior control effectiveness compared to the fixed time signal control method across both datasets. Taking the average vehicle delay in the real road network as an example, models 2–9 show reductions of 21.25%, 19.36%, 23.69%, 22.81%, 21.47%, 23.54%, 23.72%, and 27.99%, respectively, in comparison to the fixed time method. This indicates the evident advantages of employing DRL-based control models in addressing traffic signal issues. IQL can directly select the optimal action to adapt to the current environment based on the overall status of the road network. In theory, it can show better control performance, but the independent learning method also makes this method more likely to fall into local optimality. Therefore, compared with other DRL algorithms involving information interaction, IQL does not significantly improve the signal control performance of road networks. The model convergence speed is relatively slow (as shown in Fig. 7). However, due to the relatively simple network structure, the average training time of IQL is shorter (see Fig. 6).

DDQN-PER has made enhancements to the fundamental elements and model structure of the RL framework, and its control effectiveness aligns closely with IQL. In terms of model convergence, the settings of dual networks and experience replay mechanisms have improved the stability of DDQN-PER in action value function learning to a certain extent, hence the model converges slightly faster than IQL. MGMQ enhances the agent feature extraction process using DDQN, leverages GAT to aggregate lane information within intersections, and employs GraphSAGE to capture spatiotemporal information from adjacent intersections, thereby providing rich information for agent action selection. Consequently, the algorithm demonstrates significantly superior control performance compared to most baseline methods. Taking the average delay per vehicle as an example, in real data, under MGMQ, the delay is 3.10% less than that under IQL, 5.37% less than that under DDQN-PER, 1.15% less than that under CoLight, and 2.82% less than that under MN_Light. However, when compared with the value decomposition method of partial global optimal control, MGMQ still falls short in control efficacy. Furthermore, MGMQ incorporates a parameter-sharing mechanism, enabling each agent to learn from the experiences of others. This approach mitigates the need for each agent to commence learning from scratch, thereby

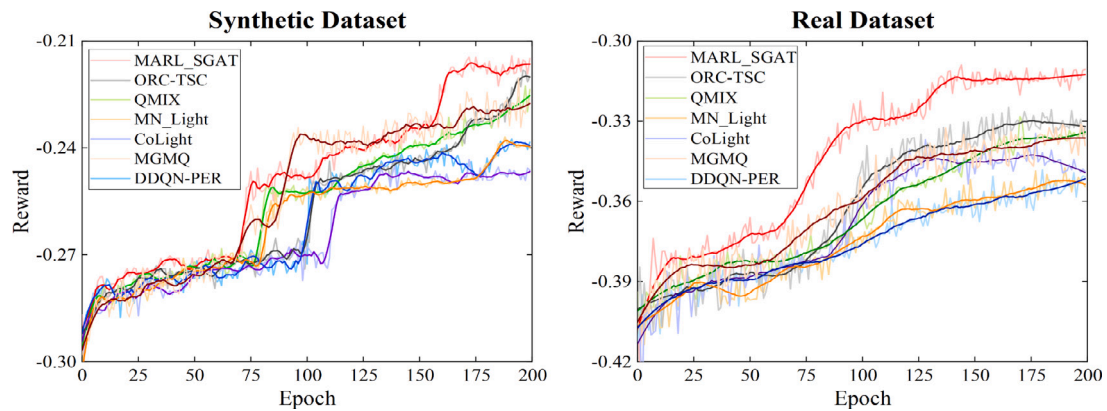


Fig. 7. Average reward during training process.

enhancing learning efficiency and reducing computational costs. CoLight and MN_Light consider the spatiotemporal correlations in road network traffic flows, utilizing GAT and bidirectional LSTM networks to extract spatiotemporal dependencies in traffic flow data, resulting in improved control over road network traffic flow. However, during the algorithm design process, the agent only selects the optimal action based on the individual action-value function. Despite the introduction of GAT and LSTM increasing agent communication, the optimal choices based on individual action-value functions still do not equate to optimal choices for the global road network, leaving a possibility for further optimization of their control effectiveness, and the convergence of the two is also at a medium level.

In comparison to the preceding five model methods, QMIX exhibits superior control effectiveness, compared with MN_Light, the signal control method based on QMIX can increase the throughput of the road network while reducing the average vehicle delay by 2.57s (real data set). This is due to the special structural design of QMIX. On the one hand, QMIX captures the deep connections between road network features by incorporating GRU units into agent networks, allowing the agent to more accurately describe the environmental state and then select actions. On the other hand, using hybrid networks, QMIX evaluates the joint action function value as a complex nonlinear combination of each agent's action function value and imposes structural constraints to maintain optimal consistency between global and individual policies. The design of consistency constraints allows the agent to select the optimal action suitable for the overall environment based only on the local environment state, reducing the state dimension in complex environments. However, the increase in the number of network layers still inevitably increases the training time of the model. QRC-TSC shares a similar control concept with QMIX but assigns a dedicated communication network to each signal agent. Based on this network, the semaphore agent can flexibly select message recipients and exchange variable-length characteristic messages with them, which makes QRC-TSC emerge as the optimal model method among the baseline methods across both datasets.

In this study, we design a state representation that comprehensively encapsulates the actual condition of the intersection and formulate a reward function that is capable of precisely measuring the impact of actions. Through an amalgamated network, employing GAT-GRU, we identify the spatiotemporal dependencies within traffic flow data. Furthermore, we simplify the search process for the optimal action-value function through a decomposition structure within the double dual network, ensuring the consistent selection of the global optimum for the road network and the individual optimum for the intersection. This accelerates network convergence. Therefore, in comparison to other baseline methods, our approach exhibits superior performance in evaluation metrics such as vehicle delay and speed. To more effectively illustrate the optimization performance of each model, we present the training rewards for some models as Fig. 7.

Analyzing the above figures and tables reveals a significantly superior optimization control performance of MARL_SGAT compared to other baseline methods. This advantage is particularly prominent in real data sets. In the real data set, even compared to the optimal method QRC-TSC among the baseline methods, MARL_SGAT still demonstrates a 5.59% reduction in average vehicle delay, a 7.59% decrease in the number of stops, a 6.39% increase in average vehicle speed, and a 1.41% increment in the number of vehicles served by the road network. In the synthetic data set, relative to QRC-TSC, MARL_SGAT exhibits a 3.5% reduction in average vehicle delay, a 5.97% decrease in the number of stops, a 5.83% increase in average vehicle speed, and a marginal 0.46% rise in the number of vehicles served by the road network. This is because there is relatively little room for improvement in the road network signal control performance under the synthetic data set. Based on the content in Section 4.1, we know that the synthetic data set corresponds to a homogeneous road network composed of nine intersections, and its traffic conditions, including flow and turning ratio, are relatively regular and stable. Since it is not derived from a real road network, the actual signal timing scheme is not included in the synthetic data set. Therefore, the initial signal timing scheme under this data set is the fixed signal timing scheme calculated by Webster's algorithm. Webster's algorithm has better control performance in a stable traffic environment (Araghi et al., 2015), hence the timing scheme under initial conditions can achieve better control performance. Under such conditions, it is reasonable that the optimization effects of each model are not obvious.

Regarding model convergence, there is no significant difference in the convergence algebra and average convergence time of MARL_SGAT and QRC-TSC. Both networks are essentially value decomposition networks that consider individual-global consistency

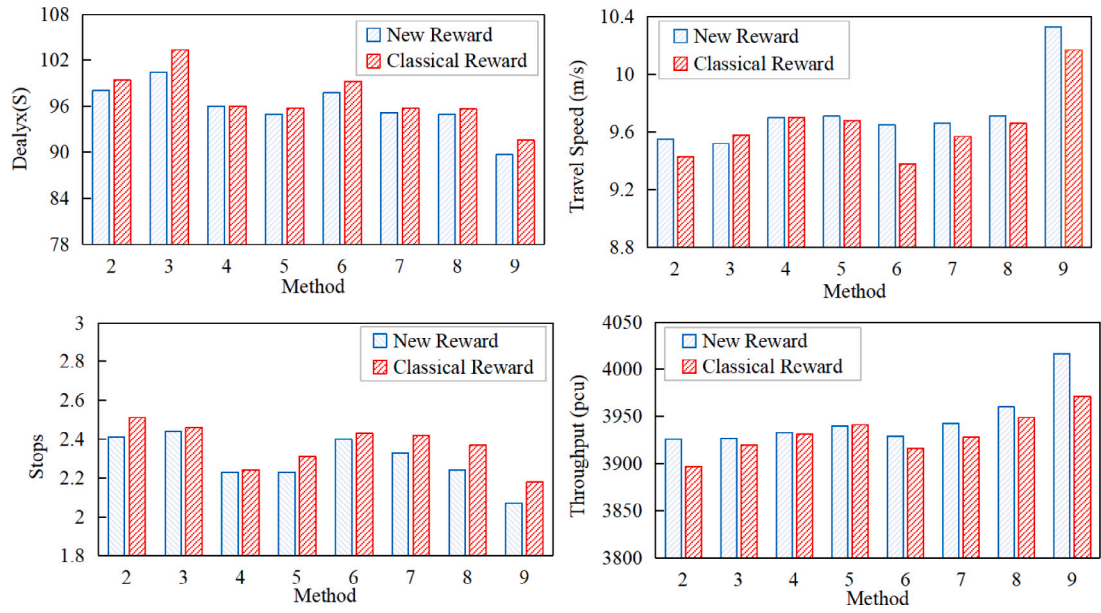


Fig. 8. Performance of algorithms under the new reward function and old reward function in the real dataset.

Table 4
Performance of ablation models.

Algorithm	Synthetic dataset			Real dataset		
	Delay (s)	Travel speed (m/s)	Stops	Delay (s)	Travel speed (m/s)	Stops
MARL_SGAT	69.72	10.17	1.89	89.68	10.33	2.07
None_ Feature extraction	74.79	9.52	2.05	97.15	9.61	2.21
None_GAT	73.89	9.21	2.10	96.79	10.26	2.22
None_GRU	74.29	9.18	2.08	96.68	10.52	2.24

constraints. The difference is that MARL_SGAT takes advantage of the easier-to-learn advantage function consistency constraints and uses a dual dueling network to decompose the action value function to speed up network convergence; QRC_TSC directly adopts the consistency constraint of the action value function and sets up a special communication network to determine the interactive information between agents.

4.3.2. Ablation experiment

Comparison of heterogeneous reward function To further investigate the efficacy of the novel reward function introduced in this study, we construct a histogram, as depicted in Fig. 8 (utilizing data from the real data set). The blue bars illustrate the model's performance under our reward function, while the orange ones reflect the simulation outcome from Wei et al. (2018) which relies on a straightforward combination of traditional reward functions. The traditional metrics used here encompass easily accessible traffic parameters such as queuing time, queuing length, or average vehicle delay. As illustrated in the figure, most model methods exhibit superior control performance under our proposed reward function. This improvement can be attributed to the reward function's design. In real road networks, intersections often display substantial heterogeneity, necessitating distinct evaluation criteria for signal agents to address different environments. The reward function introduced in this study enables each agent to holistically assess the impact of action execution on its intersection environment in the current state. This assessment is based on structural information and traffic flow data from the agent itself, as well as information from upstream and downstream intersections. Subsequently, agents can then generate the most suitable signal control policy.

Comparison of the attention modules of the space-time graph Based on the MARL_SGAT model, we separate the spatiotemporal feature extraction module, GAT, and GRU modules from the model individually. The control performance of each model is assessed then on road network traffic flow using both synthetic and real data sets. The outcomes are presented in Table 4. A substantial decline in the model's control effectiveness can be observed from Table 4 after the removal of the feature extraction module. For instance, with the real dataset, the None_Feature Extraction model exhibits an 8.11% increase in average vehicle delay, a 6.97% decrease in average vehicle travel speed, and a 6.76% rise in the number of stops compared to MARL_SGAT. Upon reintroducing the GAT and GRU modules separately, improvements in vehicle traffic situations are observed. Relative to not using the feature extraction module at all, the average number of vehicle stops in the road network decreases by 0.45% and 1.36%, respectively. While the average vehicle delay reduction is marginal at 0.17% and 0.28%, these models demonstrate a notable increase in the average vehicle speed of the

Table 5
Model convergence comparison.

Convergence index	Algorithm	
	S_MARL	MARL_SGAT
Convergent algebra	143	126
Average training time (s)	103.60	99.50

road network by 6.76% and 9.47%, respectively. This outcome to some extent validates the significance of capturing spatial and temporal dependencies in traffic flow to enhance the control efficacy of signalized intersections.

Comparison of double dual network Table 5 illustrates the convergence of the loss function in two scenarios: MARL_SGAT (where the individual value function and the advantage function are fitted by separate networks) and S_MARL (where only the individual action-value function is fitted by a separate network, and the individual state value function and advantage function are derived from the action-value function) (Wang et al., 2016). The table reveals that the convergence speed of MARL_SGAT surpasses that of S_MARL: MARL_SGAT achieves convergence in 126 generations, while S_MARL requires 143 generations for full convergence. This discrepancy arises because the decomposition structure of the dual dueling network in MARL_SGAT aids the agent in identifying the factors contributing to the optimal action-value function. This diminishes the reliance on actions during the training process of the state value function, consequently expediting the network convergence to some extent. However, the disparity in average training time between the two methods is not prominent. Specifically, the average training time of S_MARL is 103.6 s, while that of MARL_SGAT is 99.5 s, a mere 4.1 s shorter than S_MARL. The increase in network depth will result in a certain degree of computational overhead for the model.

5. Conclusion

In this research, we address the challenge posed by the considerable heterogeneity observed among adjacent intersections within urban road networks, termed “intersection heterogeneity”, encompassing variations in intersection geometric structure, traffic demand, signal timing design, and other features. To tackle the complex collaborative control problem arising from such heterogeneity, we introduce an innovative methodology named MARL_SGAT. Leveraging both authentic and synthesized datasets derived from real-world intersections, we conduct a comparative analysis of MARL_SGAT against seven distinct signal control methods. Results substantiate the superior performance of MARL_SGAT, manifesting itself through a substantial reduction in average vehicle delay and an enhancement in traffic efficiency in the road network. Remarkably, the model exhibits robust control performance in heterogeneous and homogeneous road network environments, with its advantages notably pronounced in heterogeneous settings. Furthermore, through ablation experiments, noteworthy improvements are observed in the operational conditions of road networks across various model methods following the integration of the heterogeneous reward function proposed in this study. Remarkably, MARL_SGAT demonstrates a 5.31% reduction in the average number of stops under real-world data scenarios, in contrast to instances without using the proposed reward function, resulting in the high alleviation of frequent vehicle starts and stops.

Nevertheless, it is important to acknowledge the limitations of this paper. Specifically, we have not accounted for the possible existence of special situations such as indirect left turns, U-turns, and variable lanes in the road network during the design process of the agent, potentially affecting the universality of this work. Developing MARL signal control models suitable for specific environments is a future research direction for us to explore. Additionally, our focus will also include investigating the applicability of various MARL algorithms, including but not limited to COMA and MADDPG, in large-scale signal control problems to further address collaborative traffic signal control challenges.

CRedit authorship contribution statement

Yiming Bie: Writing – original draft, Supervision, Methodology, Conceptualization. **Yuting Ji:** Writing – original draft, Visualization, Methodology, Conceptualization. **Dongfang Ma:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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